

“DSP-RF technique doubles the efficiency yet halves the size of a wireless device”



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As a signal passes through a traditional wireless device, it sees the analogue front-end, the amplifier, the mixer and an EGA that is also analogue. All these analogue components draw considerable power while bulking up your device. What if you could swap it all out with a technology that takes care of all this in one black box?

Kartik Sridharan, founder and vice president-engineering, Orca Radio Systems, spoke to Ashwin Gopinath of EFY about the advantages and challenges of using DSP-RF platform in a wireless device

What is your R&D team working on right now?

Everything from the high-speed sampling technology to the very high-speed analogue-to-digital converters and then, on the transmit side, digital power-amplifiers, up-converters and open-loop synthesisers. All of these, and DSP-RF, were developed at our India office.

Can you shed some light on DSP-RF?

DSP-RF refers to a method of communication over radio waves where we bring in DSP techniques into the radio device. This is, in essence, a traditional RF implemented using analogue techniques because signal off the air is essentially analogue, even if you are transmitting digital signals. So the front-end processing is all analogue even today.

In a traditional receiver, the first thing that the signal sees is an amplifier, which is also analogue. Then there is a mixer and an EGA, both of which are analogue. What we do is, right after the LNA, we sample the signal and perform key signal processing functions in the digital domain where

DSP techniques are brought in. That's what is different about this technology.

For design engineers, what are the benefits of using this technique?

From their perspective, they get a black box. The function of the RF is quite simple. The input is a signal from the antenna and the output is something that goes into a digital baseband processor. What happens in between is the function of the RF. The customers are not going to be enthralled just because we bring in so much digital circuitry into the device. They have to see some benefits and the benefits we bring to them are in size and power. Using DSP-RF technique, our customers get almost a 25-50 per cent improvement in power efficiency numbers as well as in size.

What was the biggest technical challenge that your design team faced with DSP-RF? How was it tackled?

Let me talk about sampler design for DSP-RF. The sam-